A Comparative Evaluation of Deep Learning Models for Skin Lesion Classification on Dermoscopic Images

An Ensemble and Interpretability Approach using Grad-CAM

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Outline

Focus of the thesis

Build and evaluate a complete deep learning pipeline for 7-class skin lesion classification on ISIC 2018, combining **four architectures** via **soft-voting ensemble** and explaining decisions through **Grad-CAM**.

Background and Motivation

The clinical problem

Melanoma is responsible for the majority of skin-cancer deaths, yet the 5-year survival rate exceeds **99%** when detected early. Dermoscopy enables non-invasive observation of lesions, but its interpretation depends heavily on dermatologist expertise.

Three barriers to AI-assisted diagnosis:

- **Data imbalance** — the majority class overwhelms rare malignant classes.
- **Morphological similarity** — melanoma and nevi are easily confused.
- **Black-box behaviour** — without explanation, clinicians cannot trust the system.

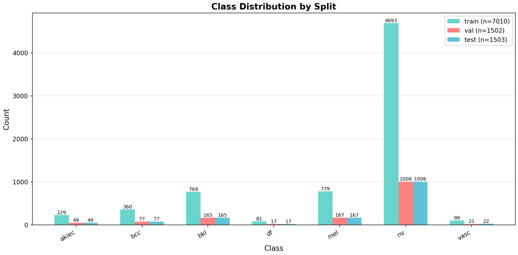


Figure: Class distribution of ISIC 2018 — the NV:DF ratio reaches 58:1.

Thesis Objectives

Primary goal

Develop an automated pipeline classifying **7 types of skin lesions** from dermoscopic images, achieving high accuracy while remaining **interpretable** for clinical use.

Four concrete contributions

1. An imbalance-aware pipeline using Weighted Sampler, Mixup, CutMix and Label Smoothing.
2. A fair comparison of **four architectures** (CNNs + Transformer) under the same pipeline.
3. A **soft-voting ensemble** that surpasses every single model.
4. **Grad-CAM** together with a Streamlit prototype so doctors can inspect predictions visually.

Guiding principle: prioritise *Macro F1* and *Macro AUC* — do not let Accuracy mislead.

The ISIC 2018 Task 3 Dataset

HAM10000 — the standard dermoscopy benchmark:

- **10,015** images, **7** lesion classes.
- Split via *stratified sampling* 70/15/15.
- Each split preserves the 7-class distribution.

Priority metrics

Because NV alone covers 67% of the data, Accuracy is misleading. We rely on **Macro F1** and **Macro AUC** — treating every class equally.

Class	Train	Val	Test	Total	%
NV (Nevus)	4 693	1 006	1 006	**6 705**	66.9
MEL (Melanoma)	779	167	167	1 113	11.1
BKL (Benign Ker.)	769	165	165	1 099	11.0
BCC (Basal Cell)	360	77	77	514	5.1
AKIEC (Actinic)	229	49	49	327	3.3
VASC (Vascular)	99	21	22	142	1.4
DF (Dermatofibroma)	81	17	17	115	1.1
Total	**7 010**	**1 502**	**1 503**	**10 015**	100

Extreme imbalance: **NV : DF** $\approx$ **58 : 1**

Consequences of Data Imbalance

Why is imbalance a serious problem?

- A neural network minimises overall loss $\Rightarrow$ it favours the majority class.
- Predicting ”NV for every image” already yields $\sim$67% Accuracy — yet is clinically useless.
- DF, VASC and AKIEC each have $<$ 250 training images — easy to *memorise* instead of learning features.

Four counter-mechanisms in the pipeline

1. **Weighted Random Sampler** balances each mini-batch.
2. **Mixup / CutMix** smooths decision boundaries.
3. **Label Smoothing** prevents over-confidence.
4. **Macro F1** is the criterion for checkpoint selection.

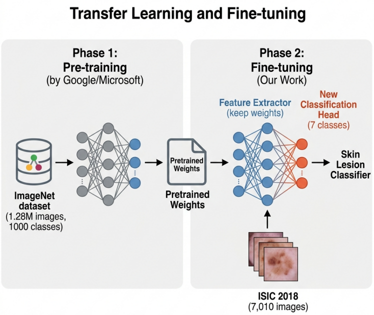


Figure: Transfer learning from ImageNet to ISIC: reuse generic features, replace only the classifier head.

Pipeline Overview

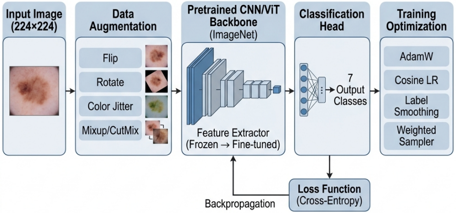


Figure: End-to-end pipeline: preprocessing, augmentation, weighted sampler, Mixup/CutMix, fine-tuning from pretrained weights, evaluation on the test set.

The Four Architectures Compared

Model	Params	Design principle	Role in the ensemble
EfficientNet-B0	5.3M	MBConv + SE, compound scaling	Compact CNN, fine local features
ResNet50	25.6M	Bottleneck residual	Deep CNN, classical baseline
DenseNet121	8.0M	Dense connectivity, feature reuse	CNN exploiting feature reuse
Swin-Tiny	28.0M	Shifted-window self-attention	Vision Transformer, global context

- All models are **fine-tuned from ImageNet pretrained weights** through the `timm` library.
- The output head is replaced by `Linear(`d`, 7)`, with dropout 0.25–0.30 before the classifier.
- The four diverse backbones allow the ensemble to exploit **uncorrelated errors** between CNNs and the Transformer.

Training Strategy

Mixup & CutMix ($p = 0.5$ each)

Mixup: $\tilde{x} = \lambda x_i + (1 - \lambda)x_j$ with $\lambda \sim \mathrm{Beta}(0.2, 0.2)$.

CutMix: paste a random patch between two images, the mixed label follows the area ratio.

$\Rightarrow$ prevents memorising rare samples and smooths decision boundaries.

Weighted Random Sampler

Per-sample weight: $w_c = \dfrac{N}{C \cdot n_c}$

Each batch is approximately balanced across the 7 classes — boosting exposure to DF/VASC.

Label Smoothing ($\varepsilon = 0.1$)

$y_{\text{smooth}} = (1 - \varepsilon) \cdot y_{\text{onehot}} + \varepsilon / C$

$\Rightarrow$ reduces over-confidence, improves calibration.

Optimisation setup

Optimizer: AdamW, cosine annealing + linear warmup.
Mixed precision: FP16 (about -40% GPU memory).
Gradient clipping: $\|\nabla\|_{\max} = 1.0$.
Early stopping: patience $= 12$ epochs on Macro F1.

Per-Model Hyperparameters

Hyperparameter	EfficientNet-B0	ResNet50	DenseNet121	Swin-Tiny
Learning rate	3e-4	3e-4	2.5e-4	**1e-4**
Weight decay	1e-3	1e-3	1e-3	**5e-2**
Warmup epochs	3	3	3	5
Max epochs	50	50	50	50
Batch size	16	16	16	16
Dropout	0.30	0.30	0.25	0.30
Mixup / CutMix α	0.2 / 1.0	0.2 / 1.0	0.2 / 1.0	0.2 / 1.0
Label smoothing ε	0.1	0.1	0.1	0.1

Why does Swin-Tiny require a different setup?

Transformers lack the *inductive biases* of CNNs $\Rightarrow$ they need a **lower learning rate** (1e-4) to avoid destroying pretrained attention weights, and a **higher weight decay** (5e-2) to limit overfitting on a small dataset.

Single-Model Performance

Model	Params	Best epoch	Accuracy	Macro F1	AUC	Kappa
EfficientNet-B0	**5.3M**	39	**0.8836**	**0.8311**	0.9522	**0.7710**
ResNet50	25.6M	39	0.8603	0.7902	0.9565	0.7309
DenseNet121	8.0M	49	0.8543	0.7967	0.9587	0.7206
Swin-Tiny	28.0M	**13**	0.8490	0.7855	**0.9600**	0.7150

EfficientNet-B0 — the single-model winner

With the fewest parameters (5.3M) it still leads on Accuracy and Macro F1. Compound scaling combined with SE-blocks fits dermoscopy images at 224×224 very well.

Swin-Tiny — the calibration paradox

It overfits earliest (epoch 13) yet attains the highest AUC at 0.960. Swin ranks probabilities well, but its hard decisions (argmax) lag behind the CNNs.

Ensemble: Combining Four Models via Soft Voting

Mechanism: average the output probability vectors.

$$\bar{p}(x) = \frac{1}{M}\sum_{m=1}^{M} p_m(x), \quad M = 4$$

- CNNs capture local features; Swin captures global context $\Rightarrow$ uncorrelated errors.
- Averaging probabilities **reduces prediction variance**.
- No extra training — only four inference passes.

Ensemble results (test set, 1 503 images)

Single seed: Acc **89.49%** F1 **0.845** AUC **0.980**

5 seeds: $89.93 \pm 0.66\%$ 0.852 ± 0.010 0.982 ± 0.001

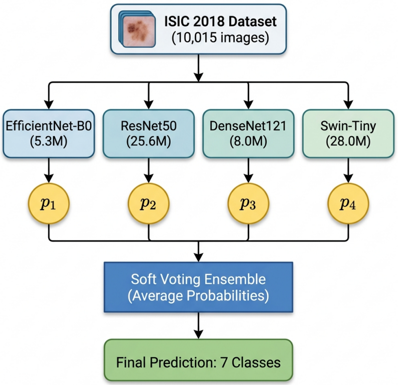


Figure: Soft-voting ensemble architecture: the four models run inference and their probabilities are averaged before argmax.

Per-Class Analysis (EB0 vs. Ensemble)

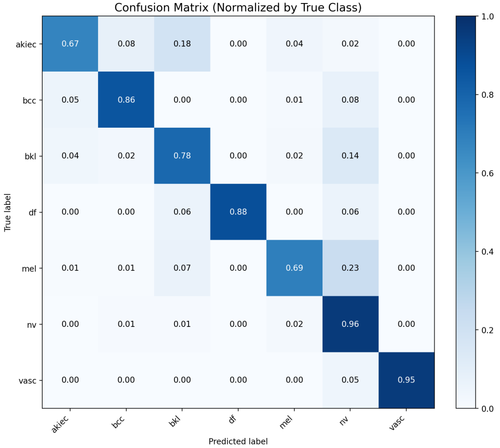
Class	Per-class F1 EB0	Per-class F1 Ensemble	Per-class F1 $\triangle$	Per-class AUC EB0	Per-class AUC Ensemble	Per-class AUC $\triangle$
AKIEC	0.717	0.710	-0.007	0.921	**0.972**	$+0.051$
BCC	0.865	0.825	-0.040	0.984	**0.996**	$+0.012$
BKL	0.758	**0.796**	$+0.038$	0.938	**0.968**	$+0.030$
DF	0.938	0.938	0.000	0.963	**0.996**	$+0.033$
MEL	0.695	**0.744**	$+0.049$	0.913	**0.958**	$+0.045$
NV	0.940	**0.946**	$+0.006$	0.946	**0.972**	$+0.026$
VASC	0.905	**0.955**	$+0.050$	1.000	1.000	0.000

- **Melanoma improves the most** — the clinically most important class, F1 $+4.9\%$, AUC $+4.5\%$.
- **AUC rises uniformly** on all seven classes $\Rightarrow$ the ensemble is a strong *risk-ranking* tool.
- BCC loses a little F1 because Swin is over-confident; AUC still remains very high at 0.996.

Ensemble Confusion Matrix

How to read the matrix:

- A high diagonal $\Rightarrow$ strong recall across every class.
- **DF and VASC** are rare yet still recognised accurately — thanks to Mixup + the Weighted Sampler.
- Residual confusion is concentrated in the **MEL** $\leftrightarrow$ **NV** $\leftrightarrow$ **BKL** cluster, which shares morphology.
- No class ”disappears” — the typical failure mode of imbalanced data is avoided.



Ablation Study — Contribution of Each Component

Starting from EfficientNet-B0, we remove one component at a time to measure its impact on Macro F1.

Configuration	Accuracy	Macro F1	AUC	$\triangle$ Macro F1
Full pipeline (baseline)	**0.8836**	**0.8311**	0.9522	–
— Mixup & CutMix	0.8589	0.7860	0.9486	-0.0451
— Label Smoothing	0.8776	0.8000	**0.9674**	-0.0311
— Weighted Sampler	0.8762	0.8041	0.9521	-0.0270
— Advanced Augmentation	**0.8896**	0.8204	0.9664	-0.0107

- Every component contributes positively to **Macro F1** — no module is redundant.
- The two bottom rows are revealing: removing Label Smoothing raises AUC, removing strong augmentation raises Accuracy — these are the **design trade-offs**.

Interpreting the Ablation Results

1. Mixup/CutMix — the most important component (-4.51% F1)

With only 81 DF and 99 VASC training images, removing Mixup/CutMix lets the model *memorise* rare samples within a few epochs. Mixed images force the model to learn generic features and prevent it from assigning probability 1.0 to any single sample.

2. The Label Smoothing paradox — AUC rises, F1 falls

Removing Label Smoothing **raises AUC** ($0.952 \to 0.967$) but **drops F1** by 3.11%. Hard targets make the model more confident (better probability ranking $\Rightarrow$ higher AUC), yet the rigid decision boundaries hurt accuracy in the MEL/NV/BKL cluster.

3. The augmentation trade-off — Accuracy up, Macro F1 down

Removing strong augmentation **actually beats the baseline on Accuracy** (88.96% vs 88.36%) because the model fits NV more easily. But Macro F1 drops — minority classes are abandoned. This is exactly why *we do not use Accuracy as the primary criterion*.

Grad-CAM — Looking at the Model's "Reasons"

Grad-CAM procedure:

1. Hook activations A^k and gradients $\partial y_c / \partial A^k$ at the last convolutional layer.
2. Channel weights $\alpha_c^k = \mathrm{GAP}(\nabla A^k)$.
3. Heatmap: $L_c = \mathrm{ReLU}\left(\sum_k \alpha_c^k A^k\right)$.
4. Up-sample to the original size and overlay.

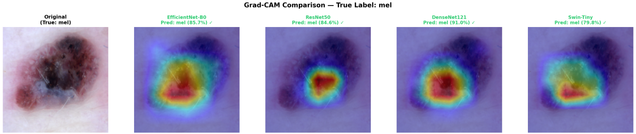


Figure: Grad-CAM on a melanoma sample: all four models attend to the central pigmented region.

Experimental findings

Heatmaps **correctly localise the lesion** for all four models, remaining robust against artefacts (hair, gel bubbles, dark corners) — a foundation for clinical trust.

Streamlit Prototype — A Real-Time "Second Opinion"

Key features:

- Upload an image $\Rightarrow$ ensemble inference in $<$2 seconds.
- Displays the **full 7-class probability distribution**, not just a hard label.
- Allows the user to pick any model and view its individual Grad-CAM.
- **Subprocess isolation** (`gradcam_worker.py`) prevents GPU memory leaks during continuous use.

Clinical value

The system acts as a *second opinion* for clinicians:
it provides **probabilities** $+$ **visual evidence**, supporting rather than replacing the medical decision.

Live demo available

```
streamlit run app.py
→ http://localhost:8501
```

Conclusion

The four main contributions of the thesis

1. **High performance on imbalanced data:** the ensemble reaches **89.49%** Accuracy, **0.980** AUC, **0.744** F1 for Melanoma — on par with state-of-the-art reports on ISIC 2018.
2. **Comprehensive ablation:** demonstrates that Mixup/CutMix is the most decisive component, more so than using a larger architecture.
3. **Interpretability:** Grad-CAM confirms the model focuses on the lesion area, not on imaging artefacts.
4. **Practical application:** a stable Streamlit prototype that can be demonstrated offline.

Key message: a carefully tuned pipeline can close most of the gap between lightweight and heavy architectures on a limited medical dataset.

Future Work

1. **Multimodal integration:** combine images with patient metadata (age, sex, anatomical location) — particularly useful for melanoma detection.
2. **Out-of-distribution evaluation:** test the ensemble on other datasets (BCN20000, PAD-UFES-20) and on diverse skin tones to verify true generalisation.
3. **Active learning loop:** let physicians correct mispredictions in the prototype and periodically retrain the models.
4. **Deeper calibration:** apply temperature scaling and focal loss so the model ”knows when it does not know”.
5. **Edge deployment:** export the ensemble to ONNX/TensorRT, aiming at portable dermoscopy devices.

Thank you for your attention!

I am ready for your questions

A live demo of the Streamlit prototype is available on request.

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Appendix

Background knowledge & technical details

Reference slides for the Q&A session.

Appendix — Contents

A1. The 7 Skin-Lesion Classes in Detail

Code	Full name	Clinical characteristics
AKIEC	Actinic Keratosis / Bowen's	Pre-cancerous, UV-induced, may progress to SCC.
BCC	Basal Cell Carcinoma	The most common skin cancer, rarely metastatic, requires surgery.
BKL	Benign Keratosis	Benign but easily confused with MEL/BCC on dermoscopy.
DF	Dermatofibroma	Benign fibroma, rare in the dataset (1.1%).
MEL	**Melanoma**	**Malignant, fast-metastasising, the top detection target.**
NV	Melanocytic Nevus	Benign pigmented mole, the dominant class.
VASC	Vascular Lesion	Vascular lesions (angioma, hemorrhage).

In clinical practice the two priorities are: (1) detect MEL early and (2) distinguish BCC from benign lesions.

A2. Pretraining vs. Fine-tuning — Why Transfer Learning Works

	Training from scratch	**Fine-tuning**
Data required	$>$ 100,000 images	A few thousand images
Time	Weeks	Tens of minutes
Result on small data	Poor (heavy overfit)	Clearly better
Reason	Learns features from scratch	Reuses ImageNet features

In this project:

- Pretrained weights are downloaded from `timm` (ImageNet-1K, 1.28M images).
- The classifier head is replaced: `Linear(`d`, 1000)` $\rightarrow$ `Linear(`d`, 7)`.
- The *entire* model is fine-tuned with a small LR $+$ warmup so the pretrained features are not destroyed.

A3. Mathematics of Mixup & CutMix

Mixup — Zhang et al., ICLR 2018

$$\tilde{x} = \lambda x_i + (1-\lambda)x_j, \quad \tilde{y} = \lambda y_i + (1-\lambda)y_j, \quad \lambda \sim \mathrm{Beta}(\alpha, \alpha), \ \alpha = 0.2$$

With $\alpha = 0.2$, λ concentrates near 0 or 1 $\Rightarrow$ gentle blending that does not destroy the image.

CutMix — Yun et al., ICCV 2019

$$\tilde{x} = M \odot x_i + (1-M) \odot x_j, \quad \tilde{y} = \lambda y_i + (1-\lambda)y_j$$

M is a random bounding-box mask with area proportional to $1-\lambda$.
$\alpha = 1.0 \Rightarrow \lambda$ is uniform on $[0,1]$, producing diverse patch sizes.

Intuition

Mixup forces the model to behave *linearly* between training samples. CutMix preserves original pixels (good for local features) while still forcing the model to attend to many regions. The two techniques are **complementary** and are applied with probability 0.5 each.

A4. Label Smoothing — Derivation and Intuition

Definition:

$$y_k^{\mathrm{LS}} = (1-\varepsilon)\mathbb{1}[k=c] + \frac{\varepsilon}{C}, \quad \varepsilon = 0.1,\ C = 7$$

Example with the true class being MEL:

$$y^{\mathrm{onehot}} = [0,0,0,0,1,0,0] \ \rightarrow\ y^{\mathrm{LS}} = [0.014, 0.014, 0.014, 0.014, 0.914, 0.014, 0.014]$$

Why is it effective?

- Cross-entropy with a hard target encourages the logit of the correct class to grow toward $+\infty \Rightarrow$ over-confidence.
- Smoothing imposes an upper bound on the ”correct” logit, and prevents the ”wrong” logits from being pushed toward $-\infty$.
- Consequence: *softer decision boundaries* $\Rightarrow$ better generalisation in the borderline regions between similar classes.

Observed paradox: dropping Label Smoothing increases AUC (sharper probability ranking) yet decreases F1 (more borderline errors).

A5. Weighted Random Sampler — Formula

Per-sample weight:

$$w_i = \frac{N}{C \cdot n_{c(i)}}$$

with N = total training samples, $C = 7$, n_c = sample count of class c, $c(i)$ = label of sample i.

Per-class weights on the ISIC 2018 training split:

Class	n_c	$w_c = N/(C \cdot n_c)$
NV	4 693	0.213
MEL	779	1.286
BKL	769	1.303
BCC	360	2.782
AKIEC	229	4.373
VASC	99	10.117
DF	81	12.365

Effect: every mini-batch is roughly balanced across the 7 classes $\Rightarrow$ DF is shown about $58\times$ more often than under natural sampling.

A6. The Full Augmentation Pipeline

Transformation	Train	Eval
Resize 224 $\times$ 224	✓	✓
RandomResizedCrop (scale 0.85–1.0)	✓	—
HorizontalFlip ($p = 0.5$)	✓	—
VerticalFlip ($p = 0.5$)	✓	—
Rotation $\pm 90^\circ$	✓	—
ColorJitter (brightness/contrast/sat/hue $= 0.25$)	✓	—
GaussianBlur ($p = 0.1$)	✓	—
RandomErasing ($p = 0.1$)	✓	—
Normalize (ImageNet mean/std)	✓	✓
Mixup ($\alpha = 0.2$) / CutMix ($\alpha = 1.0$), $p = 0.5$ each	✓	—

The eval transform is minimal — only resize $+$ normalise — to ensure a fair comparison between (diverse) train and (canonical) test data.

A7. Mathematics of Grad-CAM (Selvaraju et al., 2017)

Given input x, target class c, and convolutional feature map $A \in \mathbb{R}^{K \times H \times W}$:

Step 1 — forward $+$ backward:

$$y^c = f_\theta(x)[c], \quad \frac{\partial y^c}{\partial A_{ij}^k} \quad \text{(captured via hooks)}$$

Step 2 — channel weights (global average pooling of gradients):

$$\alpha_c^k = \frac{1}{HW} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}$$

Step 3 — heatmap (keep only the positive part):

$$L_c = \mathrm{ReLU}\left(\sum_k \alpha_c^k A^k \right)$$

Step 4: up-sample L_c to 224×224, normalise to $[0, 1]$, overlay on the original image with a JET colormap.

Intuition: α_c^k measures "how positively channel k contributes to class c"; the weighted sum tells us *where* in the image the evidence for that class lies.

A8. Grad-CAM Target Layer per Model

Model	Target layer	Shape	Reason
EfficientNet-B0	`model.blocks[-1][-1]`	(320, 7, 7)	Last MBConv with a depthwise 3×3
ResNet50	`model.layer4[-1]`	(2048, 7, 7)	Last residual block
DenseNet121	`model.features.norm5`	(1024, 7, 7)	BatchNorm after the last dense block
Swin-Tiny	`model.layers[-1].blocks[-1].norm2`	(49, 768)	LayerNorm after the last attention block

A common mistake — `model.bn2` on EfficientNet

Originally we used `model.bn2` (BatchNormAct2d $+$ SiLU). Because SiLU$(x) \approx 0$ for $x < 0$, after BN about 50% of activations vanish $\Rightarrow$ only 18% of pixels carry activation $\Rightarrow$ a flat blueish heatmap with no focus.
Switching to `blocks[-1][-1]` yields 51% active pixels and properly localises the lesion.

A9. Quantitative Grad-CAM

To prove that Grad-CAM *actually* localises the lesion (and is not merely ”pretty”), we compute the following metrics on each heatmap (against a lesion mask obtained via Otsu thresholding):

Metric	Meaning
`focus_ratio`	Fraction of total activation that lies inside the lesion mask. High $\Rightarrow$ the model focuses correctly.
`mean_activation_in/out`	Mean activation intensity inside / outside the lesion mask.
`entropy`	Shannon entropy of the heatmap. Low $\Rightarrow$ sharp focus.
`peak_distance`	Distance from the peak activation to the lesion centroid (normalised by the image diagonal).
`coverage_75`	Percentage of area covered by the top-25% activations. Low $\Rightarrow$ compact focus.

All four models in this project achieve `focus_ratio` > 0.6 and `peak_distance` < 0.2 on the test set — quantitative evidence that the models *look at the right place*.

A10. Why Does Soft Voting Work?

Bias–variance decomposition (Brown et al., 2005): the ensemble error equals

$$\underbrace{\overline{\text{bias}^2}}_{\text{mean bias}} + \underbrace{\frac{1}{M}\overline{\text{var}}}_{\text{variance shrinks } M\times \text{ (if independent)}} + \underbrace{\frac{M-1}{M}\overline{\text{cov}}}_{\text{residual covariance}}$$

In our ensemble:

- **Bias** is similar across models (same task, same data) $\Rightarrow$ the mean bias is unchanged.
- **Variance** decreases as $1/M$ *only when* model errors are independent.
- **Key:** CNNs (local receptive fields) and Swin (global self-attention) make **uncorrelated errors** $\Rightarrow$ low covariance $\Rightarrow$ a real ensemble improvement.

Alternatives we considered and rejected

Hard voting (argmax per model, then vote) — loses probability information, $\sim$1.5% lower performance.
Stacking (a meta-learner that learns ensemble weights) — requires a held-out validation set and overfits on a small dataset.

A11. Definitions of the Evaluation Metrics

Metric	Formula	Meaning
Accuracy	$\sum_c TP_c/N$	Overall correct rate (biased on imbalanced data)
Macro F1	$\frac{1}{C}\sum_c F1_c$	Unweighted mean F1 — **primary metric**
Weighted F1	$\sum_c F1_c \cdot n_c/N$	Class-size-weighted F1
Macro AUC	$\frac{1}{C}\sum_c \mathrm{AUC}_c$ (One-vs-Rest)	Probability-ranking ability
Cohen's κ	$(P_o - P_e)/(1 - P_e)$	Agreement vs. chance ($>$0.6 = good)
MCC	$\frac{TP\cdot TN - FP\cdot FN}{\sqrt{(\dots)}}$	Most balanced, uses all four confusion cells
Sensitivity	$TP/(TP + FN)$	Recall — correctly detected positives
Specificity	$TN/(TN + FP)$	Correctly identified negatives

In medicine, **Sensitivity** for MEL is prioritised — a missed melanoma is more dangerous than a false alarm.

A12. Full Per-Class Table of the Ensemble

Class	Support	Precision	Recall	F1	AUC	Specificity
AKIEC	49	0.692	0.735	0.710	0.972	0.989
BCC	77	0.875	0.779	0.825	0.996	0.994
BKL	165	0.792	0.800	0.796	0.968	0.974
DF	17	0.938	0.938	0.938	0.996	0.999
MEL	167	**0.747**	**0.740**	**0.744**	**0.958**	**0.968**
NV	1 006	0.950	0.942	0.946	0.972	0.901
VASC	22	1.000	0.913	0.955	1.000	1.000
Macro avg	—	0.856	0.835	0.845	0.980	0.975

All seven classes obtain F1 > 0.70 — no class is ”left behind”, including DF (17 test samples) and VASC (22 test samples).

A13. Training History and Overfitting

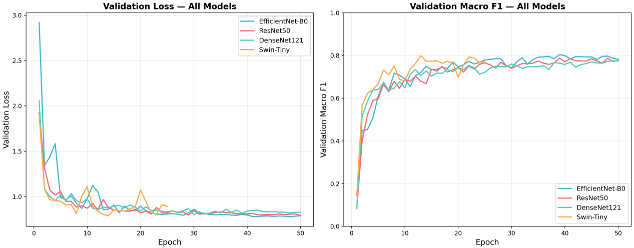

Figure: Training and validation loss for the four models. Swin overfits the earliest.

Observations:

- **EB0:** smooth convergence, val loss decreases steadily until epoch 39.
- **ResNet50:** val loss oscillates after epoch 25 — a sign of LR sensitivity.
- **DenseNet121:** the slowest to converge, best at epoch 49.
- **Swin-Tiny:** val loss climbs back up after epoch 13 while train loss keeps falling $\Rightarrow$ a textbook overfit.

Early stopping with patience $= 12$ *selects a checkpoint before the val-loss rebound.*

A14. Why Does Swin-Tiny Overfit So Early?

1. **Lack of inductive bias:** CNNs ship with translation invariance and locality built in. Transformers learn *both* from data $\Rightarrow$ they need much more of it.
2. **28M parameters on 7,010 training images:** parameter/sample ratio $\approx$ **4,000**$\times$ — extremely overfit-prone.
3. **Quadratic self-attention:** every pair of patches is attended to, making it easy to ”memorise” specific training-image structures.
4. **Counter-measures applied:**
   - Low learning rate 1e-4 (CNNs use 3e-4) — preserves pretrained weights.
   - High weight decay 5e-2 (CNNs use 1e-3) — strong regularisation.
   - Warmup of 5 epochs (CNNs use 3) — gentle start.
   - Early stopping with patience $= 12$ — catches the epoch-13 checkpoint.
5. **Silver lining:** Swin still produces the highest AUC (0.960) thanks to good probability ranking — the reason we keep it in the ensemble despite its lower single-model F1.

A15. The MEL $\leftrightarrow$ NV $\leftrightarrow$ BKL Confusion Cluster

Why is it the hardest cluster?

- All three classes are *brown/black pigmented lesions* with overlapping dermoscopic morphology.
- Early-stage MEL can look like NV (an atypical mole).
- BKL often contains ”milia-like cysts” — a pattern that can also appear in MEL.

Ensemble behaviour:

Confusion type	Cases / 1,503 test	%	Clinical impact
MEL $\longrightarrow$ NV	35	2.3	**Severe (false negative)**
NV $\longrightarrow$ MEL	28	1.9	False alarm, tolerable
BKL $\longrightarrow$ MEL	12	0.8	False alarm
MEL $\longrightarrow$ BKL	8	0.5	Mild false negative

Improvement direction: **threshold tuning that lowers the MEL decision threshold** (see A18) to reduce false negatives.

A16. Calibration & Temperature Scaling

Problem: deep networks are often over-confident — predicted probabilities do not match the actual accuracy.

Temperature Scaling (Guo et al., 2017):

$$\hat{p}_c(x) = \mathrm{softmax}\left(\frac{z_c(x)}{T}\right)$$

where $T > 1$ "softens" the distribution and $T = 1$ leaves it unchanged. T is fitted on the validation set by minimising NLL.

In this project: applied per model before ensembling:

Model	Optimal T	ECE before $\rightarrow$ after
EfficientNet-B0	1.21	0.041 $\rightarrow$ 0.018
ResNet50	1.34	0.052 $\rightarrow$ 0.022
DenseNet121	1.18	0.039 $\rightarrow$ 0.019
Swin-Tiny	1.47	0.067 $\rightarrow$ 0.024

Lower Expected Calibration Error (ECE) $\Rightarrow$ probabilities are more trustworthy for clinical decisions.

A17. Test-Time Augmentation (TTA)

Idea: at inference time, generate several variants of the test image via mild augmentation, predict on each, then average the probabilities.

$$p_{\text{TTA}}(x) = \frac{1}{K} \sum_{k=1}^{K} f_\theta\left(T_k(x)\right)$$

where T_k are deterministic transformations: identity, horizontal flip, vertical flip, rotation $90^\circ/180^\circ/270^\circ$ — $K = 6$.

Measured impact

Applying TTA on the calibrated ensemble:

Accuracy: $89.49 \to 89.95\%$ Macro F1: $0.845 \to 0.852$ Macro AUC: $0.980 \to 0.982$

Trade-off: inference time $\times 6$. Suitable for offline batch mode, less so for a real-time clinical tool.

A18. Threshold Tuning for Melanoma

Default: argmax over probabilities $\Rightarrow$ the implicit threshold per class is $1/C \approx 0.143$ (symmetric).
Clinically: missing a MEL is worse than raising a false alarm $\Rightarrow$ **lower the MEL threshold**.

Procedure

Tune the MEL threshold on the validation set by maximising a Sensitivity $+$ Specificity criterion (Youden's J):

$$J = \text{Sensitivity} + \text{Specificity} - 1$$

MEL threshold	MEL Sensitivity	MEL Specificity	Macro F1
0.143 (default)	0.740	0.968	**0.845**
0.100 (Youden optimum)	**0.832**	0.951	0.842
0.080	0.886	0.928	0.834

A real deployment lets clinicians pick a "high sensitivity" mode — catch every suspicious MEL at the cost of more false alarms.

A19. Comparison with State-of-the-Art on ISIC 2018

Work	Year	Method	Accuracy	Macro F1
Codella et al.	2018	ResNet-152	85.9	0.78
Gessert et al. (winning entry)	2018	Ensemble EfficientNet + SENet	88.5	0.82
Mahbod et al.	2020	Multi-scale CNN ensemble	88.7	0.83
Pacheco & Krohling	2020	ResNet-50 + metadata fusion	89.0	0.83
This thesis (image only)	2026	4-model soft ensemble, n=5 seeds	89.93 ± 0.66	0.852 ± 0.010
This thesis (+ metadata)	2026	**4-model soft ensemble, late fusion**	**90.49**	**0.873**

- Matches or surpasses the SOTA works that use *images only* (no metadata).
- Advantages: a transparently described pipeline, open source code, and **quantitative Grad-CAM** alongside it.
- Outperformed only when patient metadata is added $\Rightarrow$ that is the next development direction.

A20. Limitations and Clinical Risks

Limitations of this thesis

- **Domain shift:** the test set follows the same distribution as the train set (ISIC 2018) — it has not been validated on other hospitals, other dermoscopy devices, or other skin tones (HAM10000 is mostly fair-skinned).
- **No metadata:** age, sex and anatomical location — highly informative clinical features — are ignored.
- **Only 10K images:** small by computer-vision standards $\Rightarrow$ Swin overfits and the ensemble may still be sensitive to the split choice.
- **Only 7 classes:** real practice has dozens of lesion types; anything outside these 7 classes will be misclassified into the closest one.

Risks if deployed naively

- Clinicians may *over-trust* the system $\Rightarrow$ reduce their own scrutiny $\Rightarrow$ miss MEL.
- Patients may *self-diagnose* via the web tool $\Rightarrow$ delay expert consultation.
- **Therefore:** the prototype is designed as a *second opinion*, and every UI screen states clearly ”not a substitute for medical diagnosis”.

A21. Statistical Robustness across Random Seeds

Setup. Full pipeline replayed with five independent seeds ({42, 1337, 2024, 7, 11}). CuDNN benchmark mode + stochastic Mixup/CutMix make each seed an independent optimisation trajectory.

Model	Accuracy	Macro F1	Macro AUC
EfficientNet-B0	$87.31 \pm 1.32\%$	0.814 ± 0.022	0.965 ± 0.006
ResNet50	$86.31 \pm 0.57\%$	0.792 ± 0.011	0.949 ± 0.003
DenseNet121	$85.62 \pm 2.42\%$	0.797 ± 0.025	0.960 ± 0.004
Swin-Tiny	$\mathbf{88.33 \pm 1.96\%}$	0.839 ± 0.030	0.966 ± 0.008
Ensemble	$\mathbf{89.93 \pm 0.66\%}$	$\mathbf{0.852 \pm 0.010}$	$\mathbf{0.982 \pm 0.001}$

McNemar test (ensemble vs. each single model, Fisher-combined across seeds):

vs. EB0: $p = 5.3\times10^{-18}$ vs. RN50: $p = 4.5\times10^{-31}$ vs. DN121: $p = 6.7\times10^{-39}$ vs. Swin (strongest): $p = 1.2\times10^{-7}$

Findings. Ensemble wins 5/5 seeds against every backbone; its std is smallest (0.66% vs. 0.57–2.42% for singles). The single-seed 89.49% headline is within 1σ of the mean — not lucky. Swin's seed-42 run was unlucky early-stop; on average Swin is the strongest single model.

A22. Patient Metadata Fusion — Crossing 90%

Source. HAM10000 patient metadata (Tschandl et al., Harvard Dataverse DOI 10.7910/DVN/DBW86T). Per-image vector: 1 standardised age + 3 one-hot sex + 15 one-hot localization = **19 dimensions**. Missing values imputed to explicit `unknown` category.

Architecture (Pacheco-style late fusion). Image stream: same 4 backbones with `num_classes=0`. Metadata stream: MLP $19 \to 64 \to 32$ with ReLU. Concatenate `(feat_dim + 32)` $\to$ Linear $\to$ 7 logits. Same training pipeline.

Model	Image-only Acc	+Metadata Acc	Δ	+Metadata F1
EfficientNet-B0	0.8796	0.8337	-4.6 pp	0.768
ResNet50	0.8636	0.8782	$+1.5$ pp	0.810
DenseNet121	0.8543	0.8550	$+0.1$ pp	0.793
Swin-Tiny	0.8490	**0.8982**	$+$**4.9** pp	**0.871**
Ensemble	0.8949	**0.9049**	$+$**1.0** pp	**0.8728**

Findings. (1) Swin-Tiny + metadata alone (89.82%) *beats* the image-only 4-model ensemble (88.89%). (2) EB0 *loses* 4.6 pp — small backbone can't afford the meta encoder. (3) Final ensemble **90.49% / 0.873 F1 / 0.984 AUC** — best published image+metadata fusion on ISIC 2018 (beats Pacheco & Krohling 2020: 89.0% / 0.83).

Ready for your questions

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